Package 'linearAlgebra'

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Title Linear Algebra Helpers

Version 0.0.3

Description Linear algebra helpers: vectorization and half-vectorization, symmetric matrices from their half-vectorization, the duplication matrix and its Moore-Penrose inverse, deviation scores, and tests for symmetric and positive definite matrices.
Each exported function has an internal version without input checks that other projects can copy.

URL https://github.com/ijapesigan/linearAlgebra, https://ijapesigan.github.io/linearAlgebra/

BugReports https://github.com/ijapesigan/linearAlgebra/issues

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`Deviation` *Deviation from the Mean*

Description

Centers a vector, or each column of a matrix or data frame, at its mean.

Usage

```
Deviation(x)
```

Arguments

`x`	Numeric vector, matrix, or data frame.

Value

Returns a numeric matrix: one column for a vector input, the dimensions of `x` otherwise.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
Deviation(x = stats::rnorm(n = 10))
Deviation(x = matrix(stats::rnorm(n = 10 * 2), ncol = 2))
```

DMat *The Duplication Matrix*

Description

Returns the duplication matrix $\boldsymbol{D}_k$, the $k^2 \times k(k+1)/2$ matrix with $\boldsymbol{D}_k \operatorname{vech}(\boldsymbol{A}) = \operatorname{vec}(\boldsymbol{A})$ for every symmetric $k \times k$ matrix $\boldsymbol{A}$.

Usage

```
DMat(k)
```

Arguments

k	Positive integer. Dimension of the $k \times k$ matrix.

Value

Returns a numeric $k^2 \times k(k+1)/2$ matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Symmetric Functions: `PInvDmat()`, `SymofVech()`, `SymofVechs()`

Examples

```
DMat(k = 3)
```

PInvDmat *The Moore-Penrose Inverse of the Duplication Matrix*

Description

Returns the Moore-Penrose inverse of the duplication matrix, $\boldsymbol{D}_k^+ = \left(\boldsymbol{D}_k'\boldsymbol{D}_k\right)^{-1}\boldsymbol{D}_k'$, with $\boldsymbol{D}_k^+ \operatorname{vec}(\boldsymbol{A}) = \operatorname{vech}(\boldsymbol{A})$ for every symmetric $k \times k$ matrix $\boldsymbol{A}$.

Usage

```
PInvDmat(k)
```

Arguments

k	Positive integer. Dimension of the $k \times k$ matrix.

Value

Returns a numeric $k(k+1)/2 \times k^2$ matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Symmetric Functions: `DMat()`, `SymofVech()`, `SymofVechs()`

Examples

```
PInvDmat(k = 3)
```

SymofVech *Symmetric Matrix from Its Half-Vectorization*

Description

Builds the symmetric $k \times k$ matrix from $\text{vech}(\boldsymbol{A})$, its lower triangle stacked column by column. k is found from the length of `x`.

Usage

```
SymofVech(x)
```

Arguments

x	Numeric vector of length $k(k+1)/2$. Half-vectorization of a $k \times k$ matrix, $\text{vech}(\boldsymbol{A})$.

Value

Returns a numeric $k \times k$ symmetric matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Symmetric Functions: `DMat()`, `PInvDmat()`, `SymofVechs()`

Examples

```
x <- toeplitz((3:1) / 3)
x
SymofVech(x = Vech(x))
```

SymofVechs *Symmetric Matrix from Its Strict Half-Vectorization*

Description

Builds the symmetric $k \times k$ matrix from $\text{vechs}(\boldsymbol{A})$, its lower triangle without the diagonal stacked column by column, and its diagonal elements. k is found from the length of `x`.

Usage

```
SymofVechs(x, diags)
```

Arguments

`x`	Numeric vector of length $k(k-1)/2$. Strict half-vectorization of a $k \times k$ matrix, $\text{vechs}(\boldsymbol{A})$.
`diags`	Numeric vector of length 1 or k. Diagonal elements.

Value

Returns a numeric $k \times k$ symmetric matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Symmetric Functions: `DMat()`, `PInvDmat()`, `SymofVech()`

Examples

```
x <- toeplitz((3:1) / 3)
x
SymofVechs(x = Vechs(x), diags = 1)
```

`TestPositiveDefinite` *Test for a Positive Definite Matrix*

Description

Returns `TRUE` if input is a positive definite matrix, and `FALSE` otherwise.

Usage

```
TestPositiveDefinite(x, tol = 1e-06)
```

Arguments

`x`	Numeric symmetric matrix.
`tol`	Numeric. Tolerance.

Details

A $k \times k$ symmetric matrix $\boldsymbol{A}$ is positive definite if all of its eigenvalues are positive. Eigenvalues above `tol` times $|\lambda_1|$, where λ_1 is the largest eigenvalue, pass.

Value

Logical.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Wikipedia: Definite matrix

See Also

Other Linear Algebra Functions: `TestPositiveSemiDefinite()`, `TestSymmetric()`

Examples

```
# TRUE
TestPositiveDefinite(diag(2))

# FALSE
TestPositiveDefinite(
  matrix(
    data = c(1, 2, 3, 2, 4, 5, 3, 5, 6),
    ncol = 3
  )
)
```

`TestPositiveSemiDefinite`
Test for a Positive Semidefinite Matrix

Description

Returns `TRUE` if input is a positive semidefinite matrix, and `FALSE` otherwise.

Usage

```
TestPositiveSemiDefinite(x, tol = 1e-06)
```

Arguments

`x`	Numeric symmetric matrix.
`tol`	Numeric. Tolerance. An eigenvalue as small as `-tol` times $\|\lambda_1\|$ (roundoff) is accepted.

Details

A $k \times k$ symmetric matrix $\boldsymbol{A}$ is positive semidefinite if none of its eigenvalues is negative. Eigenvalues at or above `-tol` times $|\lambda_1|$, where λ_1 is the largest eigenvalue, pass. Zero eigenvalues pass, unlike in `TestPositiveDefinite()`.

Value

Logical.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Wikipedia: Definite matrix

See Also

`TestPositiveDefinite()`

Other Linear Algebra Functions: `TestPositiveDefinite()`, `TestSymmetric()`

Examples

```
# TRUE
TestPositiveSemiDefinite(x = diag(2))

# TRUE (eigenvalues 2 and 0)
TestPositiveSemiDefinite(x = matrix(data = 1, nrow = 2, ncol = 2))

# FALSE
TestPositiveSemiDefinite(
  x = matrix(
    data = c(1, 2, 3, 2, 4, 5, 3, 5, 6),
    ncol = 3
  )
)
```

TestSymmetric *Test for a Symmetric Matrix*

Description

Returns TRUE if input is a symmetric matrix, and FALSE otherwise.

Usage

```
TestSymmetric(x)
```

Arguments

x	Numeric matrix.

Details

A $k \times k$ matrix $\boldsymbol{A}$ is symmetric if $\boldsymbol{A} = \boldsymbol{A}'$. Elements are compared exactly.

Value

Logical.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Linear Algebra Functions: TestPositiveDefinite(), TestPositiveSemiDefinite()

Examples

```
# TRUE
TestSymmetric(diag(2))

# FALSE
TestSymmetric(
  matrix(
    data = 1:9,
    ncol = 3
  )
)
```

`Vec` *Vectorize a Matrix*

Description

Returns $\mathrm{vec}(\boldsymbol{A})$, the columns of a matrix stacked into one vector.

Usage

```
Vec(x)
```

Arguments

`x`	Matrix.

Value

Returns a vector of length `nrow(x) * ncol(x)`.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Vectorization Functions: `Vech()`, `Vechs()`

Examples

```
x <- toeplitz((3:1) / 3)
x
Vec(x)
```

Vech *Half-Vectorize a Matrix*

Description

Returns $\mathrm{vech}(\boldsymbol{A})$, the lower triangle of a square matrix, diagonal included, stacked column by column. If `x` has column names, the elements are named `"row,column"`.

Usage

```
Vech(x)
```

Arguments

`x`	Square matrix.

Value

Returns a vector of length $k(k+1)/2$.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Vectorization Functions: `Vec()`, `Vechs()`

Examples

```
x <- toeplitz((3:1) / 3)
x
Vech(x)
```

Vechs *Strict Half-Vectorize a Matrix*

Description

Returns $\mathrm{vechs}(\boldsymbol{A})$, the lower triangle of a square matrix, diagonal excluded, stacked column by column. If `x` has column names, the elements are named `"row,column"`.

Usage

```
Vechs(x)
```

Arguments

x	Square matrix.

Value

Returns a vector of length $k(k-1)/2$.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Vectorization Functions: `Vec()`, `Vech()`

Examples

```
x <- toeplitz((3:1) / 3)
x
Vechs(x)
```

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